Points that can talk "limit"

Chapter 3 : Limit of functions

Def 1.1 Let $A \subseteq \mathbb{R}$. A point $c \in \mathbb{R}$ is said to be a cluster point of A if for each $\delta > 0$, there exists at least a point $x \in A \setminus \{c\}$ s.t. $|x - c| < \delta$.

Some other formulations.

Let $A \subseteq \mathbb{R}$ and let $c \in \mathbb{R}$. The following statements are equivalent:

(1) c is a cluster point of A.

(2) For each $\delta > 0$, there exists at least a point $x \in A$ s.t. $0 < |x - c| < \delta$.

(3) For each $\delta > 0$, $(c - \delta, c + \delta) \cap (A \setminus \{c\}) \neq \emptyset$.

Example

(1) Let $A = [1, 3]$.

The set of cluster points of $A = A$

Pf: Let $c \in A$.

Then, for any $\delta > 0$,

If $c \neq 1$, $(c-\delta, c+\delta) \cap [1, c) \neq \emptyset$

If $c = 1$, $(c-\delta, c+\delta) \cap (1, 3] \neq \emptyset$

Hence, $c \in$ set of cluster points of A.

Let $c \notin \mathbb{R} \setminus A$.

Then, if $c < 1$, then let $\delta = 1 - c$

Then $(c-\delta, c+\delta) \cap [1, 3] = \emptyset$.

If $c > 3$, then let $\delta = c - 3$.

Then $(c-\delta, c+\delta) \cap [1, 3] = \emptyset$.

Hence, c is not a cluster point of A.

Thus, set of cluster points of $A = A$.

(2) Let $A = \{3, 3.1, 6, 7\}$.

The set of cluster points of $A = \emptyset$.

Pf: Let $c \notin A$.

Let $\delta = \inf\{|c-3|, |c-3.1|, |c-6|, |c-7|\} \neq 0$.

Then $(c-\delta, c+\delta) \cap A = \emptyset$.

Let $x \in A$.

Let $\delta = \inf(\{|c-3|, |c-3.1|, |c-6|, |c-7|\} \setminus \{0\}) \neq 0$.

Then $(c-\delta, c+\delta) \cap A = \emptyset$.

(3) Let $A = \{\frac{1}{n} : n \in \mathbb{N}\}$.

The set of cluster points of $A = \{0\}$.

Pf: Let $c = 0$.

Then, for any $\delta > 0$,

Archimedean property implies that

$\exists\ n \in \mathbb{N}$ s.t. $\frac{1}{n} < \delta$.

Then, $\frac{1}{n} \in (-\delta, \delta) \cap A \neq \emptyset$.

Let $c \notin A \setminus \{0\}$.

Let $\delta = \begin{cases} |c| & \text{if } c < 0 \\ c - 1 & \text{if } c > 1 \end{cases}$

If $0 < c \leq 1$, let $n^* \in \mathbb{N}$ s.t.

$\frac{1}{n^*+1} < c \leq \frac{1}{n^*}$.

Let $\delta = \min \{ c - \frac{1}{n^*+1}, \frac{1}{n^*} - c \}$.

One checks that $(c-\delta, c+\delta) \cap A \setminus \{c\} \neq \emptyset$.

Ex Let A be a finite subset of $\mathbb{R}$.

Then the set of cluster point of A

is $\emptyset$.

Thm 1.5 (Sequential criterion).

Let $A \subseteq \mathbb{R}$. A point $c \in \mathbb{R}$ is a cluster point of A if and only if there exists a sequence $\{a_n\}$ in A s.t.

- $\lim_{n\to\infty} a_n = c$; and
- $a_n \neq c$ for all $n \in \mathbb{N}$.

Pf: ($\Rightarrow$) Let c be a cluster point of A.

Then, by the definition of a cluster point, for any $n \in \mathbb{N}$,

$\exists\ a_n \in A \setminus \{c\}$ s.t.

$$0 < |a_n - c| < \frac{1}{n}.$$

By squeeze theorem, $\lim_{n\to\infty} a_n = c$.

Then (a_n) provides the desired sequence.

($\Leftarrow$) Let (a_n) be a sequence in A s.t.

$\lim_{n\to\infty} a_n = c$, and $a_n \neq c$ for all $n \in \mathbb{N}$.

Then, $\forall \delta > 0$,

by the definition of a limit,

$\exists N \in \mathbb{N}$ s.t.

$$|a_{N+1} - c| < \delta.$$

Combining $a_{N+1} \neq c$,

$$a_{N+1} \in (c-\delta, c+\delta) \cap A \setminus \{c\} \neq \emptyset.$$

Hence, c is a cluster point of A.

§ 1.2 Limit of functions

Def 1.6 Let $A \subset \mathbb{R}$, and let c be a cluster point of A. Let $f: A \to \mathbb{R}$ be a function.

We say that L is the limit of f at c if

$\forall\, \varepsilon > 0$, $\exists\, \delta > 0$ s.t.

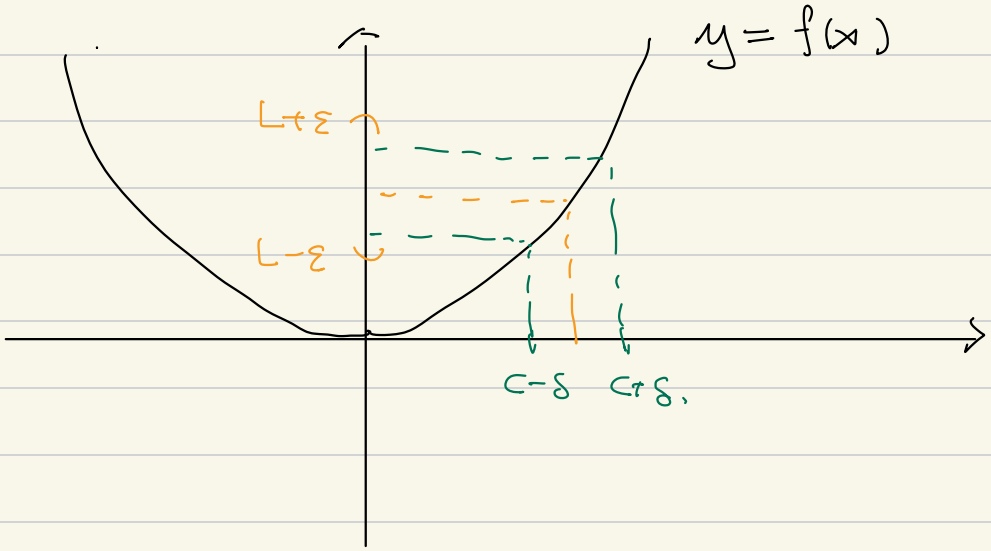
$\forall\, x \in A$ satisfying $0 < |x - c| < \delta$,

$$|f(x) - L| < \varepsilon.$$

We shall write $\lim_{x \to c} f(x) = L$.

Ex Let $f(x) = x^2$.

The curve $y = x^2$ look like:



Examples Let $b \in \mathbb{R}$. Let $c \in \mathbb{R}$

$$\lim_{x \to c} b = b.$$

Here we consider b as a constant function.

Pf: Let $f(x) = b$ for all $x \in \mathbb{R}$.

Let $\varepsilon > 0$. Choose any $\delta > 0$.

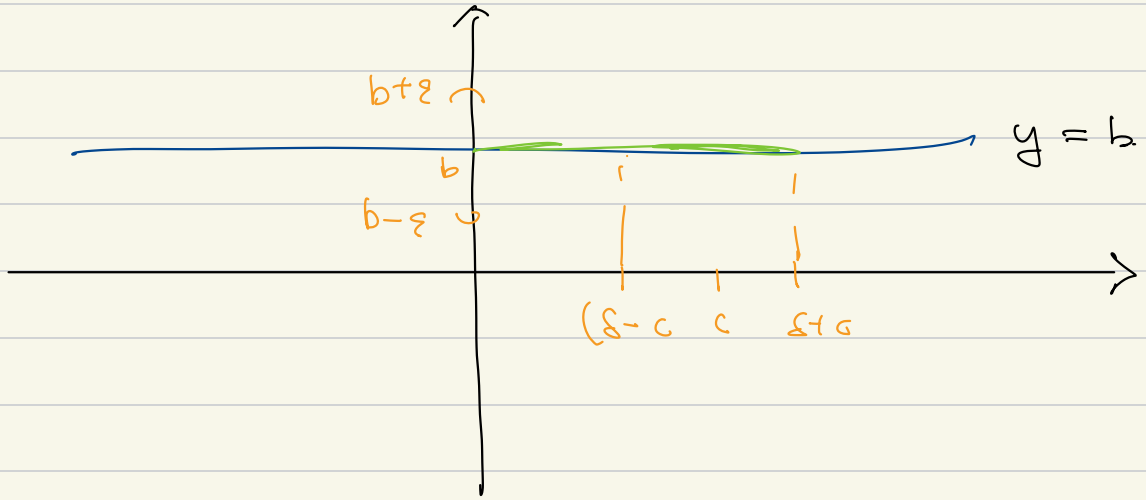
Then for any $x \in \mathbb{R}$ satisfying

$$0 < |x - c| < \delta,$$

$$|f(x) - b| = |b - b| = 0 < \varepsilon.$$

Hence $\lim_{x \to c} b = b$.

Graphical interpretation:



Ex Let $f(x) = \begin{cases} 1 & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$.

Then $\lim_{x \to 0} f = 1$.

Pf: Let $\varepsilon > 0$.

Choose any $\delta > 0$

Then for any $x \in (-\delta, \delta) \setminus \{0\}$,

$$|f(x) - 1| = |1 - 1| = 0 < \varepsilon.$$

Hence, $\lim_{x \to 0} f = 1$.

Ex $\lim_{x \to 1} \sqrt[3]{x} = 1$

Pf: Let $\varepsilon > 0$. Choose $\delta = \min\{\frac{1}{4}, \varepsilon\}$.

Consider $|\sqrt[3]{x} - 1|$

$$= \frac{|x-1|}{|\sqrt[3]{x^2} + \sqrt[3]{x} + 1|}$$

For any $x \in (1 - \frac{1}{4}, 1 + \frac{1}{4})$.

$$|\sqrt[3]{x^2} + \sqrt[3]{x} + 1| \geq \sqrt[3]{(\tfrac{3}{4})^2} + \sqrt[3]{\tfrac{3}{4}} + 1$$

$$> 1$$

Hence, for any $x \in (1-\delta, 1+\delta) \setminus \{1\}$,

$$|\sqrt[3]{x} - 1| < |x-1| < \delta \leq \varepsilon.$$

By definition of the limit,

$$\lim_{x \to 1} \sqrt[3]{x} = 1.$$

Ex Let $f: \mathbb{R} \to \mathbb{R}$ given by:

$$f(x) = \begin{cases} x & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}.$$

Then $\lim_{x \to 0} f = 0$.

Pf: Let $\varepsilon > 0$.

Let $\delta = \varepsilon$

Then, for any $x \in (-\delta, \delta) \setminus \{0\}$,

$$\begin{cases} \text{if } x \text{ is rational, then } |f(x)| = |x| < \delta = \varepsilon. \\ \text{if } x \text{ is irrational, then } |f(x)| = 0 < \varepsilon. \end{cases}$$

By definition, $\lim_{x \to 0} f = 0$.

§ 2 Sequential criterion.

Thm 2.1 Let $A \subseteq \mathbb{R}$. Let $f: A \to \mathbb{R}$ and let c be a cluster point of A.

Then the following statements are equivalent:

(i) $\lim_{x \to c} f(x) = L$

(ii) For any sequence (x_n) in $A \setminus \{c\}$ converging to c, the sequence $(f(x_n))$ converges to L.

Pf: (i) $\Rightarrow$ (ii).

Suppose $\lim_{x \to c} f(x) = L$.

Let $\varepsilon > 0$.

By the definition of a limit,

$\exists \quad \delta > 0 \quad$ s.t.

for any $x \in (c-\delta, c+\delta) \setminus \{c\}$,

$|f(x) - L| < \varepsilon$. — (*)

Let (x_n) be a sequence in $A \setminus \{c\}$

converging to c.

Then, by the definition of convergence,

$\exists\ N \in \mathbb{N}$ s.t.

$$\forall\ n > N,\ 0 < |x_n - c| < \delta.$$

and so by $(*)$,

$$|f(x_n) - L| < \varepsilon.$$

By the definition of a limit of a sequence,

$$\lim f(x_n) = L.$$

(ii) $\Rightarrow$ (i)

Suppose (i) does not hold.

$\exists\ \varepsilon > 0$ s.t. for any $n \in \mathbb{N}$

$$\exists\ x_n \in \left(c - \tfrac{1}{n}, c + \tfrac{1}{n}\right) \setminus \{c\}$$

s.t. $|f(x_n) - L| \geq \varepsilon$.

Then $\left| \lim_{n\to\infty} f(x_n) - L \right| \geq \varepsilon$.

Hence $\lim_{n\to\infty} f(x_n) \neq L$.

This contradicts (i).

Thm 2.2 (Uniqueness of a limit)

Let $A \subseteq \mathbb{R}$. Let $f : A \to \mathbb{R}$. Let c be a cluster point of A. Then there exists at most one limit of f at c.

Pf: This follows from Thm 2.1 and the uniqueness of a limit of a sequence.

§ 2.3 Divergence criterion

Thm 2.3 Let $A \subset \mathbb{R}$. Let $f: A \to \mathbb{R}$.

Let c be a cluster point of A.

If there exists two sequences (x_n) and (y_n) in $A \setminus \{c\}$ converging to c such that $\lim_{n\to\infty} x_n$ and $\lim_{n\to\infty} y_n$ exist, and $\lim_{n\to\infty} x_n \neq \lim_{n\to\infty} y_n$, then $\lim_{x\to c} f$ does not exist.

Example Let

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}.$$

Let $x_n = \frac{1}{n}$, $\forall n \in \mathbb{N}$

Then $\lim_{n \to \infty} x_n = 0$, and

$$\lim_{n \to \infty} f(x_n) = 1$$

Let $y_n = -\frac{1}{n}$, $\forall n \in \mathbb{N}$.

Then $\lim_{n \to \infty} y_n = 0$, and

$$\lim_{n \to \infty} f(y_n) = -1$$

Hence, $\lim_{x \to 0} f$ does not exist.

Example Let $f: \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by

$$f(x) = \sin\left(\frac{1}{x}\right).$$

Let $x_n = \frac{1}{2\pi n}$

Then, $\lim_{n\to\infty} x_n = 0$

$$\lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} \sin(2\pi n) = 0$$

Let $y_n = \frac{1}{2\pi n + \frac{\pi}{2}}$

Then, $\lim_{n\to\infty} y_n = 0$

$$\lim_{n\to\infty} f(y_n) = \lim_{n\to\infty} \sin\left(2\pi n + \frac{\pi}{2}\right) = 1.$$

By Thm 2, 3, $\lim_{x\to 0} f$ does not exist.

Graph:

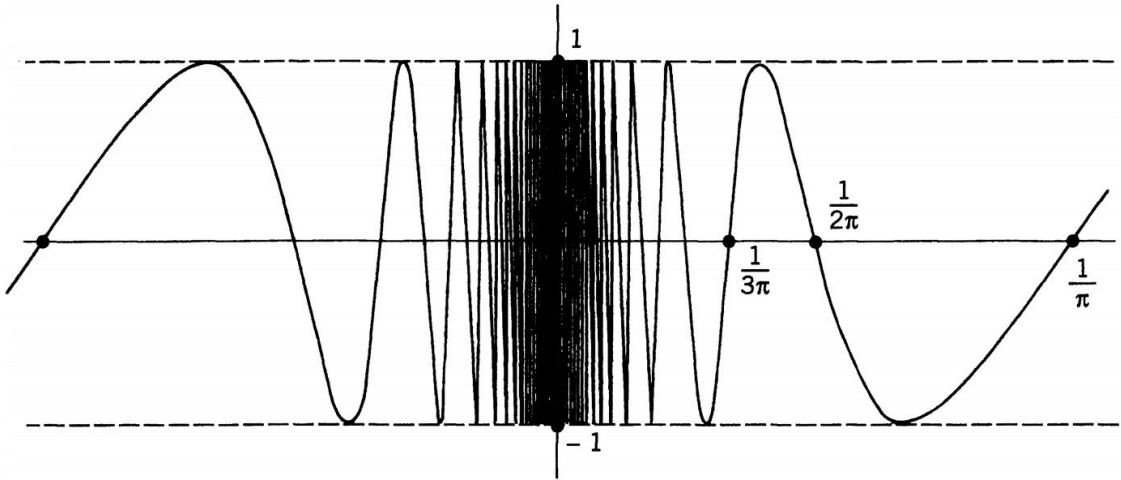


Figure 4.1.3 The function $g(x) = \sin(1/x)(x \neq 0)$

§ 3 The boundedness of a limit

Thm 3.1 (Boundedness of a limit)

Let $A \subseteq \mathbb{R}$. Let $f: A \to \mathbb{R}$.

If the limit of f at c exists,

then f is bounded on $A \cap (c-\delta, c+\delta)$

for some $\delta > 0$.

Pf: Let $L = \lim_{x \to c} f$.

Choose $\varepsilon = 1$.

By the definition of limit,

$\exists\ \delta > 0$ s.t.

$$\forall\ x \in (c-\delta, c+\delta) \setminus \{c\},$$

$$|f(x) - L| < 1$$

By $\big||f(x)| - |L|\big| \le |f(x) - L|$,

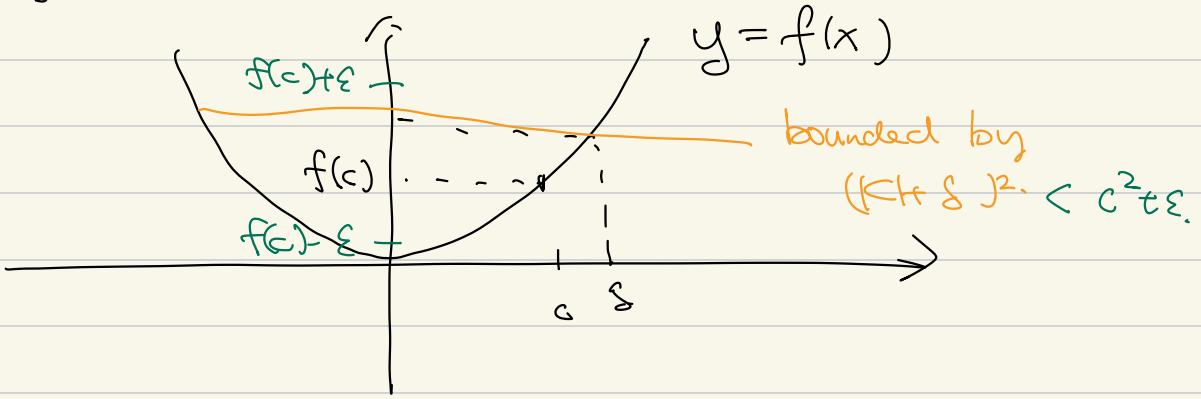
we have $|f(x)| < |L| + 1$.

Hence, $\forall\ |x-c| < \delta$,

$$|f(x)| < \sup\{|L|+1, |f(c)|\}.$$

Example

(1) Let $f(x) = x^2$.



(2) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$, $f(x) = \frac{1}{x}$.

Since f is not bounded on $(-\delta, \delta)$ for any $\delta > 0$, $\lim_{x \to 0} f$ does not exist.

(3) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} n & \text{if } x = \frac{1}{n} \text{ for some } n \in \mathbb{N} \\ 0 & \text{otherwise.} \end{cases}$$

Since f is not bounded on $(-\delta, \delta)$ for any $\delta > 0$, $\lim_{x \to 0} f$ does not exist.

§ 4 Arithmetic on limits

§ 4.1 Basic operations

Def 4.1 Let $A \subseteq \mathbb{R}$. Let $f, g : A \to \mathbb{R}$.

Define

(1) (Sum) $(f+g)(x) = f(x) + g(x)$ for all $x \in A$

(2) (Difference) $(f-g)(x) = f(x) - g(x)$ for all $x \in A$

(3) (Product) $(fg)(x) = f(x)\,g(x)$ for all $x \in A$

(4) (Multiple) Let $b \in \mathbb{R}$. $(bf)(x) = b \cdot f(x)$ for all $x \in A$.

(5) (Quotient) If $g(x) \neq 0$ for all $x \in A$,

then $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ for all $x \in A$.

Thm 4.2 Let c be a cluster point of A.

Suppose $\lim_{x \to c} f(x) = L$, and $\lim_{x \to c} g(x) = M$.

Then (1) $\lim_{x \to c} (f+g) = L + M$

(2) $\lim_{x \to c} (f-g) = L - M$

(3) $\lim_{x \to c} fg = LM$

(4) Let $b \in \mathbb{R}$. Then $\lim_{x \to c} (bf) = bM$

(5) If $g(x) \neq 0 \quad \forall x \in A$ and $M \neq 0$,

$$\lim_{x \to c} \frac{f}{g} = \frac{L}{M}.$$

Pf: Use the sequential criterion

Let (x_n) be a sequence converging to c.

Then the sequential criterion implies that

$$\lim_{n \to \infty} f(x_n) = L \quad , \text{ and } \quad \lim_{n \to \infty} g(x_n) = M.$$

Then $$\lim_{n \to \infty} (f+g)(x_n) = \lim_{n \to \infty} \left(f(x_n) + g(x_n)\right)$$

$$= \lim_{n \to \infty} f(x_n) + \lim_{n \to \infty} g(x_n)$$

$$= L + M.$$

Example Let $f : (-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \infty) \to \mathbb{R}$ given by

$$f(x) = \frac{4x^2 - 1}{2x - 1}.$$

Let $c \neq \frac{1}{2}$. Then, $\lim_{x \to c} f(x) = \frac{4c^2 - 1}{2c - 1} = 2c + 1$.

Let $c = \frac{1}{2}$. Then, $\lim_{x \to \frac{1}{2}} f(x) = \lim_{x \to \frac{1}{2}} \frac{4x^2-1}{2x-1}$

$= \lim_{x \to \frac{1}{2}} (2x+1)$

$= 2$

§ 4.2 Squeeze Theorem

Thm 4.6 (Squeeze Theorem)

Let $A \subseteq \mathbb{R}$. Let $f, g, h : A \to \mathbb{R}$, and let c be a cluster point of A.

Suppose • $f(x) \leq g(x) \leq h(x)$ for all $x \in A$ with $x \neq c$

• $\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L$.

Then $\lim_{x \to c} g$ exists and $\lim_{x \to c} g = L$.

Pf: Let (x_n) be any sequence in A.

Then, by squeeze theorem for sequences,

$\lim_{n \to \infty} f(x_n) = \lim_{n \to \infty} h(x_n) = L$ implies that

$\lim_{n \to \infty} g(x_n) = L$.

Since (x_n) is arbitrary, the sequential criterion implies $\lim_{n \to \infty} g(x_n) = L$

Exercise (a) Find $\lim_{x\to 0} x^2(1+\sin(x^2+1))$.

For any $x \in \mathbb{R}$,

$$0 \leq x^2(1+\sin(x^2+1)) \leq 2x^2$$

Since $\lim_{x\to 0} 2x^2 = 0$, the sequeeze theorem implies that $\lim_{x\to 0} x^2(1+\sin(x^2+1)) = 0$.

(b) We have the following two inequalities:

For $x \geq 0$, $x - \frac{1}{6}x^3 \leq \sin x \leq x$.

For $x < 0$, $x - \frac{1}{6}x^3 \geq \sin x \geq x$.

Then, for $x \neq 0$,

$$1 - \frac{1}{6}x^2 \leq \frac{1}{x}\sin x \leq 1$$

$$\lim_{x\to 0}\left(1 - \frac{1}{6}x^2\right) = 1$$

By Squeeze theorem, $\lim_{x\to 0} \frac{\sin x}{x} = 1$.

§5 Other limits

§5.1 One sided limits

Def 5.1 Let $A \subseteq \mathbb{R}$ and let $f: A \to \mathbb{R}$.

(1) Let c be a cluster point of $A \cap (c, \infty)$

We say that L is right hand limit of f at c, written as $\lim_{x \to c+} f = L$

if for any given $\varepsilon > 0$, there exists $\delta > 0$ s.t. for all $x \in A$ with $0 < x - c < \delta$, then $|f(x) - L| < \varepsilon$.

ensure non-empty

This is to restrict the function f to $A \cap (c, \infty)$ and then consider the limit.

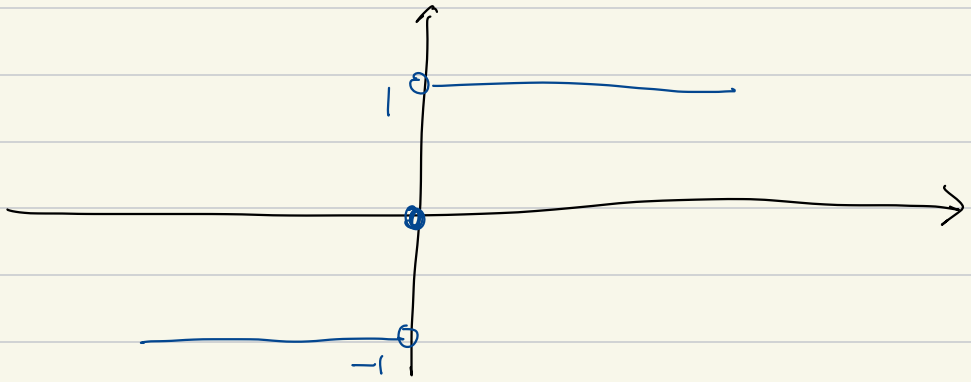
(2) Let c be a cluster point of $A \cap (\infty, c)$

We say that L is a left hand limit of f at c, written as $\lim_{x \to c^-} f = L$ if for any given $\varepsilon > 0$, there exists $\delta > 0$ s.t. for all $x \in A$ with $0 < c - x < \delta$, then $|f(x) - L| < \varepsilon$.

Thm 5.2 Let $f: A \to \mathbb{R}$. Let c be a cluster point in both sets $A \cap (c, \infty)$ and $A \cap (-\infty, c)$. Then $\lim_{x \to c} f = L$ if and only if $\lim_{x \to c^+} f = \lim_{x \to c^-} f = L$

Example Let

$$f(x) = \operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

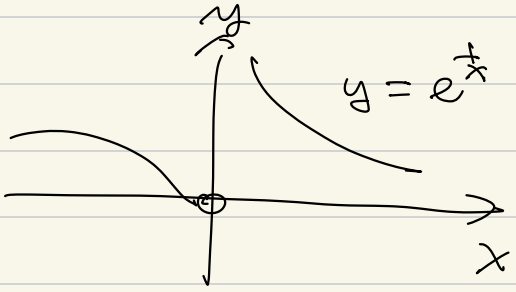


Then $\lim_{x \to 0^+} f = 1$ and $\lim_{x \to 0^-} f = -1$

Thus $\lim_{x \to 0} f = 1$ does not exist

Ex. Let $f(x) = e^{\frac{1}{x}}$ defined for $x \neq 0$.

Find $\lim_{x \to 0} e^{\frac{1}{x}}$?



We first compute $\lim_{x \to 0^+} f$.

For $x > 0$, we have the inequality

$$\frac{1}{x} < e^{\frac{1}{x}}$$

$\lim_{x \to 0^+} f$ does not exist.

We then compute $\lim_{x \to 0^-} f$.

For $x < 0$, we have

$$-\frac{1}{x} < e^{-\frac{1}{x}}$$

And so $0 < e^{-\frac{1}{x}} < -x$

As $\lim_{x \to 0^-} (-x) = 0$., by squeeze theorem,

$$\lim_{x \to 0^-} e^{-\frac{1}{x}} = 0.$$

§ 5.2 Limit tending to infinity

Def 5.5 Let $A \subseteq \mathbb{R}$ and let $f: A \to \mathbb{R}$.

Let c be a cluster point of A.

(1) We write $\lim_{x \to c} f = \infty$ if for any $\alpha \in \mathbb{R}$,

$\exists\ \delta > 0$ s.t. for any $x \in A$ with $0 < |x - c| < \delta$,

$f(x) > \alpha$.

(2) We write $\lim_{x \to c} f = -\infty$ if for any $\alpha \in \mathbb{R}$,

$\exists\ \delta > 0$ s.t. for any $x \in A$ with $0 < |x - c| < \delta$,

$f(x) < \alpha$.

Example Let $f: \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = \frac{1}{x^4}$.

For any given $\alpha > 0$, choose $\delta = \frac{1}{\sqrt[4]{\alpha}}$. Then

for any $0 < |x| < \delta$,

$$f(x) = \frac{1}{x^4} > \frac{1}{\delta^4} > \alpha.$$

Hence, $\lim_{x \to 0} f = \infty$.

§ 5.3 Limit at infinity

Condition analogous to that ∞ is a cluster pt.

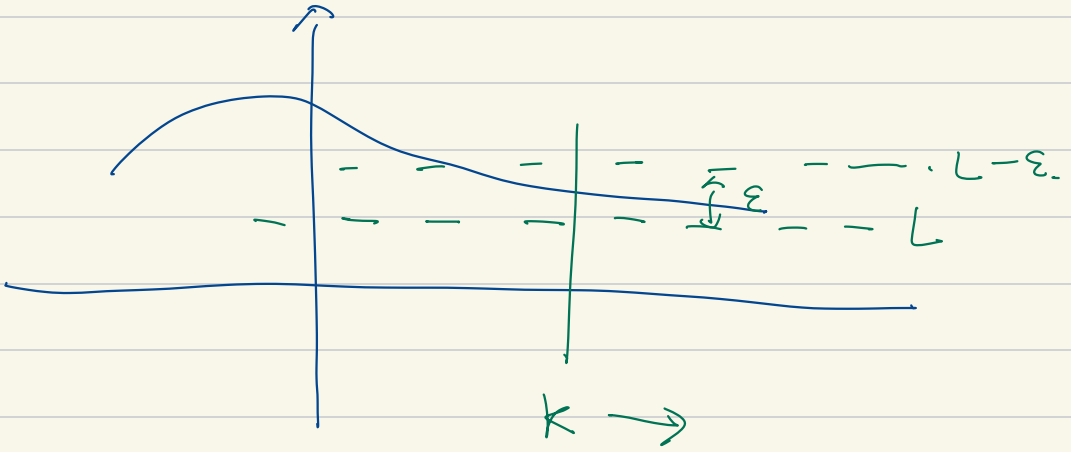
Def 5.7 Let $A \subseteq \mathbb{R}$, and let $f: A \to \mathbb{R}$.

Suppose $(a, \infty) \subset A$ for some $a \in \mathbb{R}$.

We write $\lim_{x \to \infty} f = L$ for some $L \in \mathbb{R}$

if for any $\varepsilon > 0$, there exists $K > 0$ s.t.

for any $x > K$, $|f(x) - L| < \varepsilon$.



Def Let $A \subseteq \mathbb{R}$, and let $f: A \to \mathbb{R}$.

Suppose $(a, \infty) \subset A$ for some $a \in \mathbb{R}$.

We write $\lim_{x \to -\infty} f = L$ for some $L \in \mathbb{R}$

if for any $\varepsilon > 0$, there exists $K < 0$ s.t.

for any $x < K$, $|f(x) - L| < \varepsilon$.

Example.

Let $f: \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = \frac{1}{x}$.

Then $\lim_{x \to \infty} f = 0$.

Pf: Let $\varepsilon > 0$.

Choose $K > 0$ s.t. $K > \frac{1}{\varepsilon}$.

Then, $\forall\, x > K$,

$$\left| \frac{1}{x} \right| < \frac{1}{K} < \varepsilon$$

By definition, $\lim_{x \to \infty} \frac{1}{x} = 0$.